

Underfitting and Overfitting in Machine Learning: From Bias–Variance Trade-offs to Modern Generalization Theory

Abstract

The central objective of supervised machine learning is not to minimize training error, but to achieve low expected error on unseen data drawn from the same distribution. This paper synthesizes the classical theory of underfitting and overfitting through the bias–variance decomposition, model-complexity analysis, and regularization, and then connects these ideas to modern phenomena including double descent and benign overfitting. We present a unified diagnostic framework that distinguishes high-bias from high-variance regimes, review Ridge, Lasso and Elastic Net regularization, and emphasize the practical roles of cross-validation, learning curves and careful evaluation design. The classical U-shaped error curve remains foundational for many practical settings, yet recent work shows that highly over-parameterized models can interpolate training data while still generalizing well. The deepest lesson is that effective complexity—not mere parameter count—governs generalization, and that model selection must be driven by estimates of population risk rather than empirical risk alone.

1. Introduction

When a machine learning model is trained, it is tempting to treat minimization of training error as the primary objective. The training set, however, is only a finite sample from an underlying population. The true goal is to learn a mapping that continues to perform well on new observations drawn from the same distribution. This distinction yields the fundamental inequality of supervised learning:

Good training performance $\neq$ Good generalization

Underfitting arises when the hypothesis class lacks sufficient expressive power to capture the systematic signal present in the data. Overfitting arises when the model becomes so flexible that it begins to treat sample-specific noise as if it were meaningful structure. Both phenomena are best understood through the complementary lenses of signal-versus-noise reasoning and the bias–variance decomposition of expected prediction error.

The present paper organizes classical and modern theory into a compact narrative suitable for practitioners and researchers who require a clear conceptual map rather than an exhaustive literature survey. We retain the most consequential definitions, mathematical statements, diagnostic instruments and contemporary caveats, illustrating them with schematic figures and comparative tables. The classical U-shaped risk curve is presented first; the discussion then proceeds to the modern complications of double descent and benign overfitting, the principal families of explicit regularization, and a practical diagnostic and remediation framework.

2. Signal versus Noise: The Core Distinction

A productive way to frame the entire subject is to view every training observation as the superposition of an underlying systematic relationship and stochastic noise:

$$Y = f(X) + \varepsilon, \quad \text{where } E[\varepsilon] = 0 \text{ and } \text{Var}(\varepsilon) = \sigma^2.$$

The function f is the signal we wish to recover; ε represents irreducible uncertainty arising from unmeasured factors, measurement error, or inherent stochasticity. The learning algorithm therefore faces a fundamental tension: it must extract enough of the signal to be useful, yet remain sufficiently insensitive to the particular realization of noise present in any finite training sample.

Underfitting corresponds to insufficient recovery of the signal. Overfitting corresponds to the absorption of noise into the fitted function, so that the model's predictions on new data become contaminated by idiosyncrasies of the training sample. Good generalization occupies the intermediate regime in which the important systematic structure is captured while residual noise is largely ignored. This perspective immediately suggests both diagnostic questions (Is the model missing structure, or is it fitting idiosyncrasies?) and remedial directions (increase capacity, constrain capacity, or collect more data).

3. Underfitting and Overfitting Defined

3.1 Underfitting

A model underfits when its hypothesis class—or its effective complexity after regularization—is too restrictive to represent the target relationship. The classic illustration is a quadratic truth $y = x^2$ confronted by a learner restricted to the linear family $\{a + bx\}$. No matter how large the training sample becomes, a straight line cannot reproduce curvature; the difficulty is representational rather than statistical. Underfitting is therefore characteristically associated with high bias: a systematic discrepancy that persists across repeated independent draws of training data.

In practice, underfitting may also result from an overly strong regularization penalty, from missing interaction terms or nonlinear transformations, or from an architecture whose inductive bias is simply mismatched to the problem. Collecting more data does not cure a fundamental misspecification of the hypothesis class.

3.2 Overfitting

Overfitting is the complementary pathology. A highly flexible model is able to pass nearly perfectly through every training observation, including those that deviate from the true signal solely because of noise. Training error continues to decline, yet test error rises. The fitted function exhibits high variance: small perturbations of the training sample induce large changes in the learned predictor. A high-degree polynomial fitted to a modest collection of noisy points remains the pedagogical archetype of this behavior, but the same qualitative pattern appears with deep decision trees, unregularized high-dimensional linear models, and over-parameterized neural networks trained without appropriate inductive biases or early stopping.

Table 1. Classical three-regime characterization of model complexity.

Model complexity	Bias	Variance	Typical outcome
Very low	High	Low	Underfitting
Appropriate	Moderate	Moderate	Good generalization
Very high	Low	High	Overfitting

4. The Bias–Variance Decomposition

Bias quantifies the systematic discrepancy between the average prediction of a learning procedure—taken over the distribution of possible training sets—and the true regression function:

$$\text{Bias}^2(x) = (E_D[f_D(x)] - f(x))^2$$

Variance measures how much the fitted model fluctuates when the training sample is replaced by another independent draw from the same population:

$$\text{Var}_D[f_D(x)] = E_D[(f_D(x) - E_D[f_D(x)])^2]$$

Under squared-error loss the expected prediction error admits an exact three-term decomposition (Geman, Bienenstock & Doursat, 1992):

$$E[(Y - f(X))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

The final term, σ^2 , is irreducible noise that no choice of model can eliminate. Increasing the flexibility of the hypothesis class typically reduces bias while simultaneously increasing variance, generating the classical trade-off. The optimal complexity is the point at which the marginal reduction in bias is exactly offset by the marginal increase in variance, so that total expected error is minimized. Domingos (2000) later established those analogous decompositions exist for a broader family of loss functions, confirming that the conceptual tension is not an artifact of squared error alone.

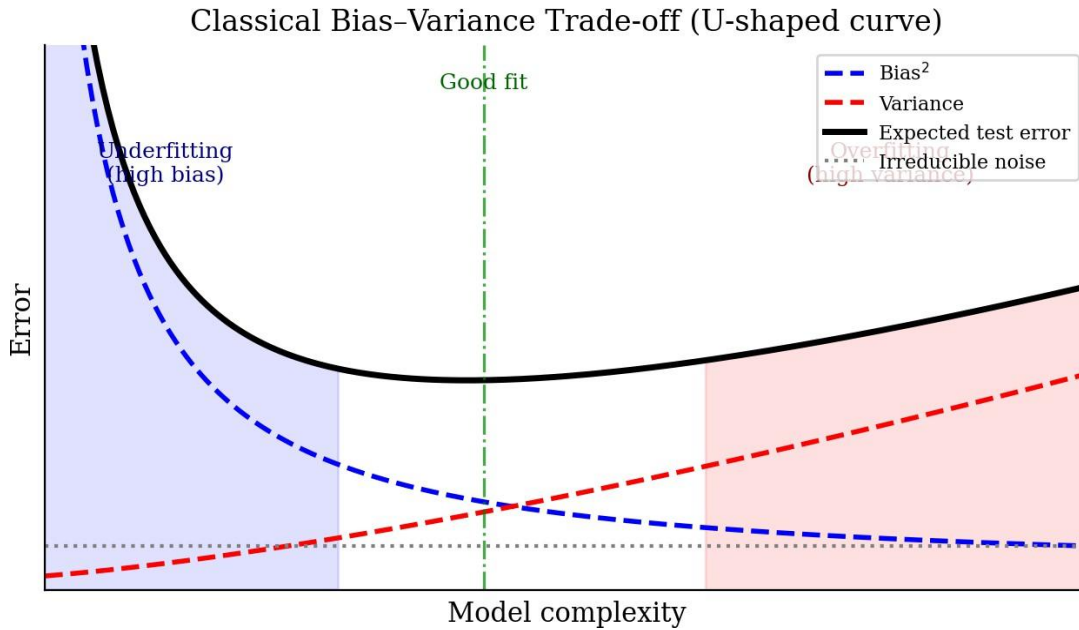


Figure 1. Classical U-shaped test-error curve arising from the bias–variance trade-off. At low complexity, bias dominates (underfitting); at high complexity, variance dominates (overfitting). Optimal performance occurs at intermediate complexity where the sum of the two contributions is minimized.

5. Empirical Risk versus Population Risk

A learning algorithm has access only to the empirical risk computed on the observed training sample,

$$R'(f) = (1/n) \sum_{i=1}^n L(y_i, f(x_i)).$$

The quantity of genuine scientific interest is the population risk,

$$R(f) = E_{(X,Y) \sim P}[L(Y, f(X))],$$

which is the expected loss under the true data-generating distribution. Overfitting is precisely the situation in which $R'(f)$ is driven very low while $R(f)$ remains large. The difference $R(f) - R'(f)$ is called the generalization gap. Any model-selection procedure that relies solely on training error is therefore optimizing the wrong objective. This distinction, emphasized throughout the statistical learning literature and restated by Belkin et al. in the context of modern interpolating methods, is the reason that independent validation and test sets are indispensable.

6. Model Complexity Illustrated by Polynomial Regression

Polynomial regression remains the cleanest controlled demonstration of the classical trade-off. One begins with a degree-1 (linear) model and successively raises the maximum degree, thereby generating a nested sequence of hypothesis classes of strictly increasing capacity. Training mean-squared error almost invariably declines monotonically with degree. Validation or test error, by contrast, typically follows a U-shaped trajectory: high at low degree (underfitting), minimized at an intermediate degree that balances bias and variance, then rising again once the polynomial begins to interpolate noise (overfitting).

Because the complexity parameter—the polynomial degree—is discrete, transparent and easy to visualize, the experiment continues to serve as a standard pedagogical and diagnostic device even in an era dominated by deep networks. Qualitatively identical patterns appear when the complexity knob is tree depth, number of radial-basis centers, or network width, provided one remains inside the classical under-parameterized or only mildly over-parameterized regime.

A practical experimental protocol for any concrete data set is therefore: fit a sequence of models of increasing capacity, record training, validation and test metrics for each, and plot the resulting curves. The location of the validation-error minimum supplies an empirical estimate of the optimal complexity for that problem and sample size.

7. Modern Complications: Double Descent and Benign Overfitting

The classical narrative asserts that once model capacity exceeds a critical threshold, test error must rise. Contemporary research has revealed additional regimes. Belkin, Hsu, Ma and Mandal (2019) demonstrated the phenomenon of *double descent*: after the classical U-shaped portion of the risk curve, test error can decrease a second time as capacity increases far beyond the interpolation threshold—the point at which the model first becomes capable of fitting the training data essentially perfectly.

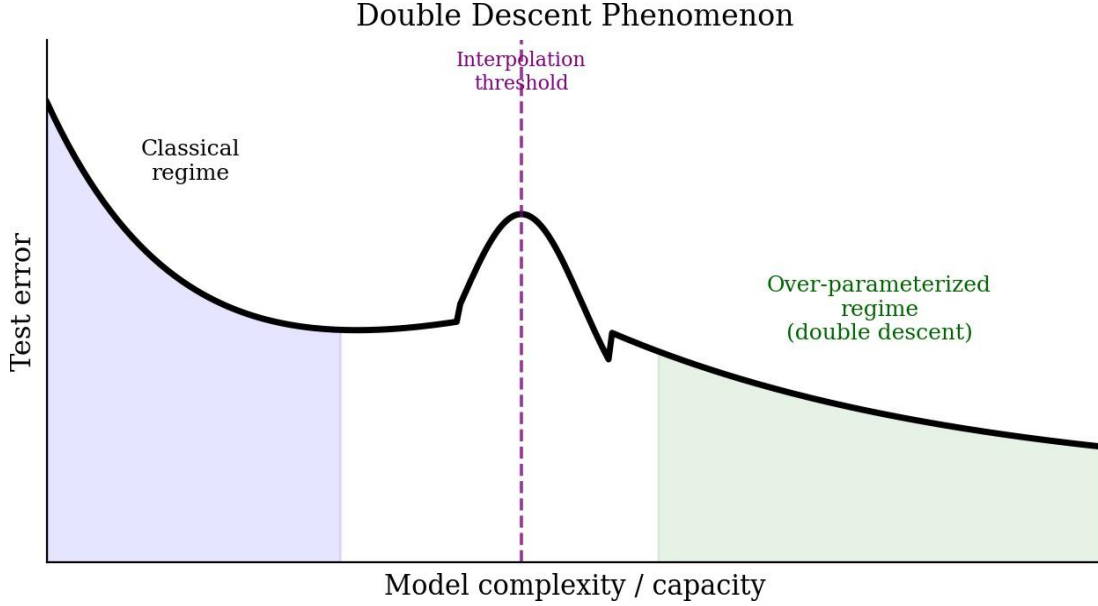


Figure 2. Schematic of the double-descent risk curve. After a peak near the interpolation threshold, further increases in capacity can again reduce test error in the heavily over-parameterized regime. The second descending limb is the distinctive signature of double descent.

A closely related concept is *benign overfitting* (Bartlett et al., 2020). There exist theoretically characterized settings—particularly in high-dimensional linear regression—in which a model interpolates the training sample, including its noise, yet still achieves near-optimal prediction performance on fresh data. Consequently, the unqualified statement “a training error of zero implies harmful overfitting” is no longer universally valid. Parameter count alone is an incomplete measure of effective complexity; architecture, the geometry of the data, the trajectory of the optimizer, and both explicit and implicit regularization all influence whether interpolation is benign or harmful.

Zhang et al. (2017) supplied a complementary empirical demonstration: modern neural networks can memorize randomly assigned training labels, proving that the capacity to interpolate does not by itself explain why the same architectures generalize when trained on structured data. The practical implication is that classical bias–variance reasoning remains the most useful first-order model for the majority of everyday tabular and moderate-capacity problems, while the behavior of heavily over-parameterized models constitutes an active research frontier that enlarges rather than discards the classical picture.

8. Regularization: Explicit Control of Complexity

When overfitting is driven by excess effective degrees of freedom, a principled remedy is to augment the empirical-risk objective with an explicit complexity penalty:

$$\text{minimize } (1/n) \sum_i L(y_i, f_{\beta}(x_i)) + \lambda \Omega(\beta),$$

where Ω measures complexity and $\lambda \geq 0$ controls the strength of the penalty. Regularization reframes the optimization problem so that, among all models that explain the training data reasonably well, solutions possessing desirable structural properties—small coefficient norm, sparsity, smoothness—are preferred.

8.1 Ridge Regression (L2 Regularization)

Ridge regression (Hoerl & Kennard, 1970) employs the squared Euclidean penalty $\sum_j \beta_j^2$. Its closed-form estimator is $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$. The added multiple of the identity matrix stabilizes the inversion of $\mathbf{X}^T \mathbf{X}$ whenever predictors

are collinear or the design matrix is otherwise ill-conditioned, and continuously shrinks coefficient magnitudes toward zero. The procedure deliberately accepts a modest increase in bias in exchange for a typically larger reduction in variance—an explicit operationalization of the bias–variance trade-off.

8.2 Lasso (L1 Regularization) and Elastic Net

The Lasso (Tibshirani, 1996) replaces the squared penalty with the L1 norm $\sum_j |\beta_j|$. The geometry of the L1 ball induces sparsity: some coefficients are driven exactly to zero, yielding simultaneous regularization and automatic variable selection. Elastic Net combines the two penalties through a convex combination controlled by a mixing parameter $\alpha \in [0,1]$, recovering pure Lasso at $\alpha = 1$ and pure Ridge at $\alpha = 0$. The hybrid is often preferred when groups of correlated predictors are present, because it tends to retain or discard entire groups together rather than arbitrarily selecting a single representative.

Table 2. Comparison of Ridge and Lasso regularization.

Property	Ridge (L2)	Lasso (L1)
Penalty term	$\sum \beta^2$	$\sum \beta $
Shrinks coefficients	Yes (continuously)	Yes
Produces exact zeros	Generally, no	Yes (sparsity)
Automatic feature selection	No	Yes
Behavior with correlated predictors	Often stable (keeps groups)	Tends to select one of a group
Closed-form solution	Yes	No (requires convex optimization)

An important practical observation is that regularization frequently *increases* training error while decreasing test error. The apparent degradation on the training sample is the necessary price of improved generalization—the concrete manifestation of the bias–variance trade-off in everyday model fitting.

9. Model Selection, Validation and Diagnostics

Because training error is an optimistically biased estimator of population risk, both model complexity and hyper-parameters must be chosen by reference to an independent estimate of generalization performance. K-fold cross-validation partitions the available data into K roughly equal folds, trains on K–1 fold, evaluates on the held-out fold, and averages the K resulting scores. After every selection decision has been finalized, a completely untouched test set supplies a final, approximately unbiased assessment. Repeated exploratory use of the same validation data gradually converts it into a de-facto training set and re-introduces optimism bias; the test set must therefore remain strictly isolated until the very end of the modeling pipeline.

Learning curves (error versus training-set size) and validation curves (error versus a complexity hyper-parameter such as polynomial degree, tree depth or regularization strength) furnish visual diagnostics of the prevailing bias–variance regime:

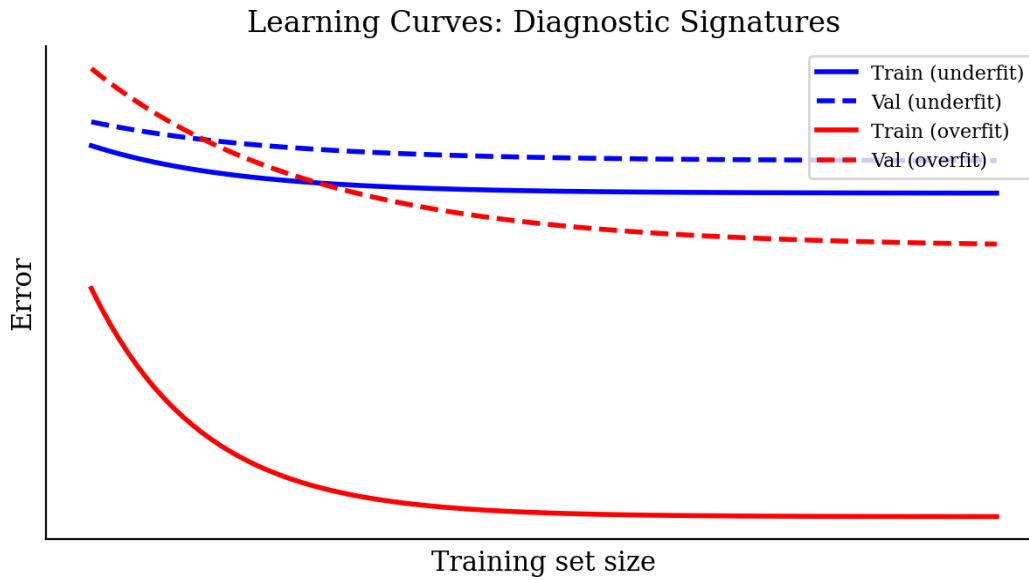


Figure 3. Schematic learning curves. Persistently high and closely spaced training and validation errors suggest underfitting (high bias). A large gap between very low training error and substantially higher validation error suggests overfitting (high variance).

Table 3. Practical diagnostic matrix.

Training performance	Validation / Test	Likely interpretation
Poor	Poor	Underfitting / high bias
Excellent	Poor	Overfitting / high variance
Good	Good	Healthy generalization
Excellent	Excellent on random split but poor in production	Possible leakage or distribution shift

10. Practical Remedies

Against underfitting. Increase model capacity (higher-degree polynomial features, deeper or wider networks, more flexible kernels); engineer additional informative predictors or interaction terms; reduce an overly aggressive regularization coefficient; or improve the representation of the problem so that the hypothesis class becomes appropriate to the underlying relationship.

Against overfitting. Reduce capacity; apply Ridge, Lasso, Elastic Net or other explicit penalties; collect additional training data (especially effective in high-variance regimes); perform feature selection or dimensionality reduction; employ early stopping monitored on a validation trajectory; or average multiple models through bagging or related ensemble methods.

More data is not a universal remedy. If the hypothesis class is fundamentally incapable of representing the target relationship, additional samples merely produce a more precise estimate of the wrong function. Likewise, an excellent validation score obtained under data leakage, target contamination, or a material train–test distribution shift does not guarantee real-world performance. These failure modes are conceptually distinct from classical overfitting and must be diagnosed by separate checks on the data-generation and evaluation pipeline.

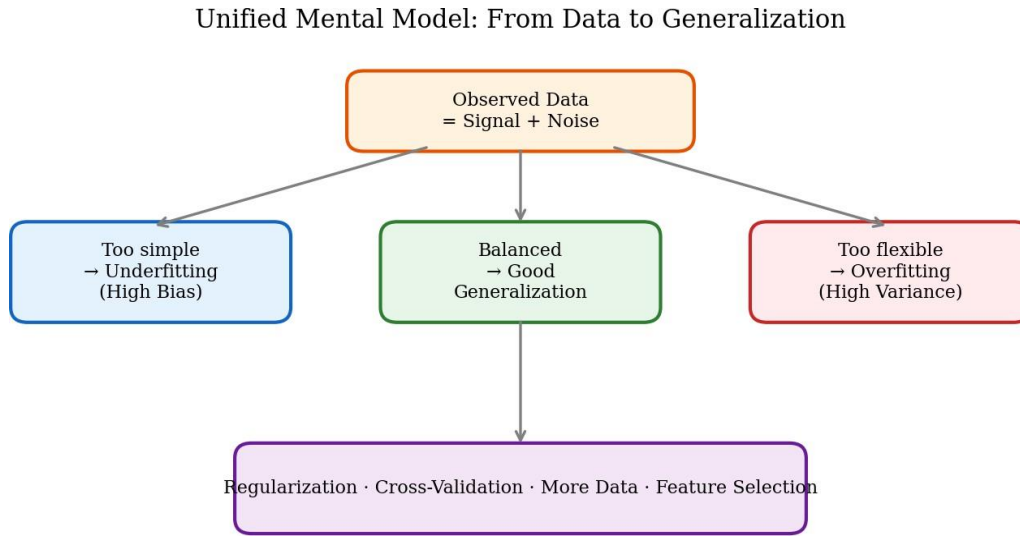


Figure 4. Unified conceptual map linking observed data (signal + noise), the three classical complexity regimes, and the principal levers—regularization, cross-validation, additional data, and feature selection—that improve generalization.

11. Capacity, Stability and Effective Complexity

Statistical learning theory formalizes the question of why a predictor that performs well on a finite sample should also perform well on unseen data. One classical capacity measure is the Vapnik–Chervonenkis (VC) dimension, which quantifies the richest collection of binary labelings a hypothesis class can shatter. High-capacity classes can realize many patterns—including pure noise—raising the risk of overfitting unless the sample size grows correspondingly. A simplified generalization-bound intuition takes the form

$$R(f) = R'(f) + \text{Complexity penalty}(n, \text{capacity}),$$

where the additive penalty shrinks with sample size n and grows with the effective capacity of the hypothesis class. The precise constants and functional forms depend on the loss and on the particular capacity measure employed; the expression should be read as conceptual guidance rather than a universal numerical formula.

Algorithmic stability supplies a complementary perspective (Bousquet & Elisseeff, 2002). If the removal or replacement of a single training point changes the learned predictor only modestly, generalization bounds can be derived that do not rely on explicit combinatorial capacity measures such as VC dimension. Stability, variance and regularization are therefore intimately related: methods that dampen sensitivity to individual observations tend both to reduce variance and to enjoy stronger generalization guarantees.

In modern deep learning the notion of *effective complexity* has become central. Two networks that possess identical parameter counts can generalize very differently depending on architecture, initialization scale, optimization trajectory and the implicit bias of the training algorithm. Parameter count is therefore an incomplete proxy; the geometry of the particular solution that is found matters at least as much as the nominal size of the hypothesis class.

12. Discussion

The classical bias–variance picture remains the most useful first-order mental model for the majority of practical supervised-learning problems, especially those involving tabular data, moderate sample sizes and models whose capacity is not extreme. Double descent and benign overfitting enlarge rather than discard that picture: they identify additional operating regimes in which the relationship between capacity and risk is non-monotonic. The correct practical stance is therefore nuanced. Classical diagnostics and regularization remain essential everyday tools, while awareness of modern phenomena prevents overly rigid conclusions of the form “more parameters must imply worse generalization.”

Equally important is the recognition that poor generalization can arise from causes other than classical overfitting. Label noise, data leakage, covariate shift, concept drift, and evaluation protocols that inadvertently reuse information can all produce the outward appearance of a model that “fails to generalize.” A complete diagnostic process therefore examines training-versus-validation gaps, the shapes of learning and validation curves, stability under data perturbation, and the integrity of the entire data-generation and splitting pipeline.

Finally, the hierarchy of evaluation must be kept in view: training fit is not the same as generalization to a random test split, and generalization to a random test split is not the same as sustained performance under real-world distribution shift. Responsible model development attends to all three layers.

13. Conclusion

Underfitting reflects a failure to extract sufficient signal from the training sample; overfitting reflects the extraction of non-transferable sample idiosyncrasies. The classical bias–variance decomposition supplies a precise mathematical language for this tension under squared loss, while regularization, cross-validation and capacity control supply the principal engineering responses. Modern theory has enlarged the landscape by revealing double descent and benign overfitting, demonstrating that interpolation of the training data need not be fatal. The enduring principle, however, remains unchanged: a machine-learning model is ultimately judged by the quality of the structure it transfers to data it has never seen. Selecting, training and evaluating models according to estimates of population risk—rather than empirical risk alone—is the practical embodiment of that principle.

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